CITS5508 Machine Learning
Introduction to Machine Learning

Marcell Szikszai

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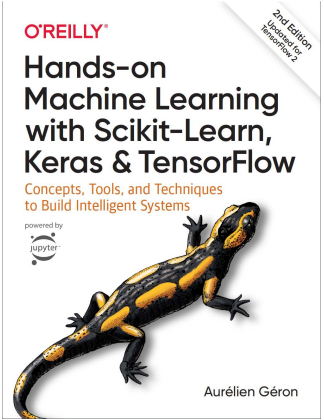
THE UNIVERSITY OF WESTERN AUSTRALIA

The Machine Learning Landscape

- What is Machine Learning?
- Why use Machine Learning?
- Types of Machine Learning Systems
- Main Challenges of Machine Learning
- Testing and Validating

Reading



Chapter 1 – The Machine Learning landscape

Hands-on Machine Learning with Scikit-Learn & TensorFlow,
Aurélien Géron,
O'Reilly Media, 2022.

Available online through UWA OneSearch.

Example code and solutions to the exercises:
https://github.com/ageron/handson-ml3/blob/main/01_the_machine_learning_landscape.ipynb

What is machine learning (ML)?

When might you encounter ML on a regular day?

- When your email services decide an email is spam, what is it doing?
- When Spotify suggests the next song to you, what is it doing?

What is machine learning (ML)?

ML algorithms are tools for the automatic acquisition of knowledge.

ML is also known as inductive learning.

Inductive learning is a form of logical inference that allows you to obtain generic conclusions about a *particular set of examples.*

What is machine learning (ML)?

Machine Learning is the science (and art) of programming computers so they can learn from data.

> *"[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.'*
>
> *-Arthur Samuel, 1959*

> *"A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."*
>
> *-Tom Mitchell, 1997*

Example – spam filter

Let's identify these parts in the spam filter.

Experience E:

- examples of spam emails (e.g. flagged by users)
- examples of regular/non-spam emails

Task T:

- Decide if a new email is spam or not.

Performance measure P:

- How good is my filter to identify new emails as spam or non-spam?
  - The performance measure P needs to be defined and will depend on the problem.
  - In this example, accuracy–the ratio of correctly classified emails–is one possible performance measure.

Example – spam filter

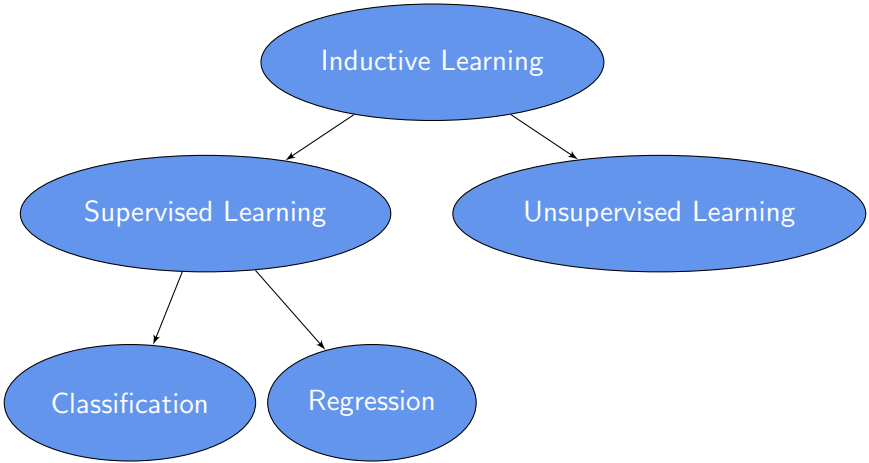
	Descriptive features				Target feature
ID	`word_freq_all`	`word_freq_internet`	`word_freq_sale`	$\cdots$	Outcome
1	0.5	0.8	0.7	$\cdots$	Spam
2	0.1	0.5	0.9	$\cdots$	Spam
3	0.2	0.2	0.3	$\cdots$	Regular
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
n	0.3	0.4	0.1	$\cdots$	Regular

`word_freq_`{`word`}: percentage of words in the e-mail matching {`word`}.

- Each example/instance is described by a feature vector (or attribute vector) and by the associated label.
- The objective of the learning algorithm is to construct a classifier that can correctly assign the label of new (unlabelled) examples.

Based on the data set available at https://archive.ics.uci.edu/dataset/94/spambase.

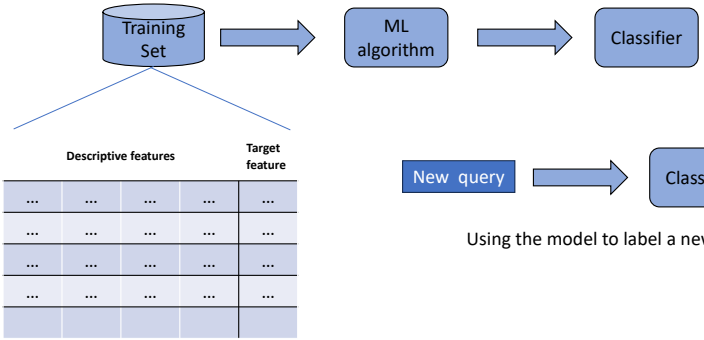
Training Supervision



We will start by looking at **supervised** approaches.

Supervised learning

Learning a model from a set of historical examples



Using the model to label a new example

Supervised learning – some definitions

Attribute: Also called features, predictors, independent variables. An attribute describes a characteristic or aspect of an example.

Example: Also called instance, register, data point. It is a tuple of attribute values that describes an object of interest (e.g., a regular email, a patient, or a company's customer history).

Label: Also called the target or dependent variable. It is a special attribute that describes the phenomenon of interest, i.e., the concept that the ML algorithm should learn to make predictions about it. For classification, the target variable is typically a nominal label set $\{L_1, L_2, \ldots, L_k\}$. For regression, it is a real value.

Supervised learning – some definitions

Data set: A data set is composed of examples with respective attribute values and the associate label. Below is a data set $\mathcal{D}$ containing n examples and m attributes.

	X_1	X_2	$\cdots$	X_m	Y
D_1	x_{11}	x_{12}	$\cdots$	x_{1m}	y_1
D_2	x_{21}	x_{22}	$\cdots$	x_{2m}	y_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
D_n	x_{n1}	x_{n2}	$\cdots$	x_{nm}	y_n

Row D_i refers to the i-th example. That is, $i = 1, 2, \ldots, n$.

The entry x_{ij} refers to the j-th value of attribute X_j for example D_i.

Examples are tuples $D_i = (x_{i1}, x_{i2}, \ldots, x_{im}, y_i) = (\vec{x_i}, y_i)$ also denoted as (x_i, y_i), where it is implied x_i is a vector.

Supervised learning – example

Let's use another example: credit approval. The bank wishes to automate the process of evaluating credit card applications, as it receives thousands daily.

What would the examples be?

- Customer personal information, such as annual salary, years in residence, outstanding loans, etc.

How can the bank create a target feature?

- Each customer record also contains whether approving credit for that customer was a good idea.

Supervised learning – example

The input $\vec{x}$: the customer information that is used to make a credit decision.

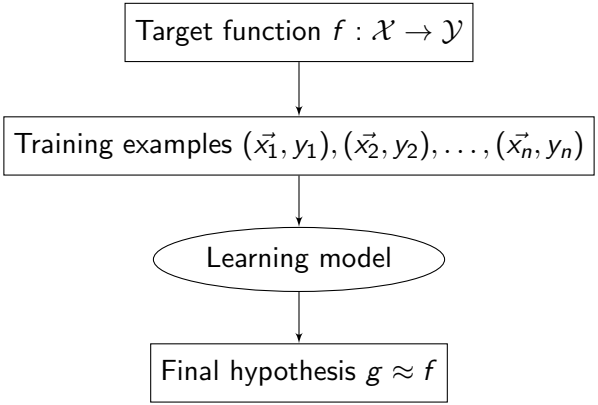
The desired unknown function $f : \mathcal{X} \to \mathcal{Y}$: ideal function for credit approval, where $\mathcal{X}$ is the input space (set of all possible inputs $\vec{x}$), and $\mathcal{Y}$ is the output space (set of all possible outputs).

The data set $\mathcal{D}$ contains all input-output examples, $(\vec{x}_i, y_i)$.

The learning algorithm uses the data set $\mathcal{D}$ to find a function $g : \mathcal{X} \to \mathcal{Y}$ that approximates f.

The function g is selected from a set of candidate functions under consideration, referred to as the hypothesis space $\mathcal{H}$. For instance, $\mathcal{H}$ could represent the set of all linear functions that the algorithm can select.

Components of learning



The learning model is formed by the learning algorithm and hypothesis set $\mathcal{H}$.

Supervised learning – simple classification model

Consider $\mathcal{X} = \mathbb{R}^m$ as the input space, where $\mathbb{R}^m$ is the m-dimensional Euclidean space.

Let $\mathcal{Y} = \{-1, +1\}$ be the output space (binary classification).

In the credit approval example, the input vector $\vec{x} \in \mathbb{R}^m$ contains the attributes in a credit application, such as salary, years in residence, outstanding debit, etc.

The binary output y defines the decision, approving or denying credit.

The hypothesis space $\mathcal{H}$ is specified through a functional form shared by all hypothesis $h \in \mathcal{H}$.

Supervised learning – simple classification model

Our functional form $h(\vec{x})$ gives different weights $(w_1, w_2, \ldots, w_m)$ to the different coordinates in $\vec{x}$. This represents the relative importance of the attributes in the credit decision.

$$\text{Credit is approved if} \quad \sum_{i=1}^{m} w_i x_i > \text{threshold},$$

$$\text{Credit is not approved if} \quad \sum_{i=1}^{m} w_i x_i < \text{threshold}.$$

Supervised learning – simple classification model

Our simple model can be rewritten as,

$$h(\vec{x}) = \operatorname{sgn}\left(\left(\sum_{i=1}^{m} w_i x_i\right) + b\right),$$

where $x_1, \ldots, x_m$ are the components of the vector $\vec{x}$, the weights $w_1, w_2, \ldots, w_m$, and the bias term b are trainable parameters. $\operatorname{sgn} x$ is the sign function,

$$\operatorname{sgn} x := \begin{cases} -1 & \text{if } x < 0, \\ +1 & \text{if } x > 0. \end{cases}$$

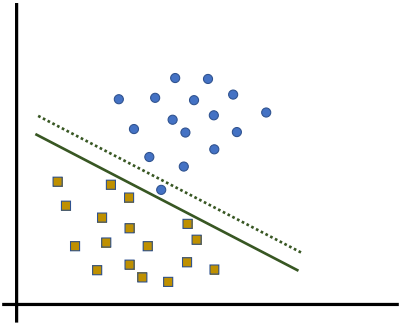
$h(\vec{x}) = +1$ means 'credit approved' and $h(\vec{x}) = -1$ means 'credit denied'.

Supervised learning – simple classification model

Our weights $w_1, w_2, \ldots, w_m$, and the bias term b determines the threshold as the bank will approve credit. Credit is approved if

$$\sum_{i=1}^{m} w_i x_i > -b.$$

Our model or hypothesis $g \in \mathcal{H}$ will be determined by the optimal choices of weights and bias.



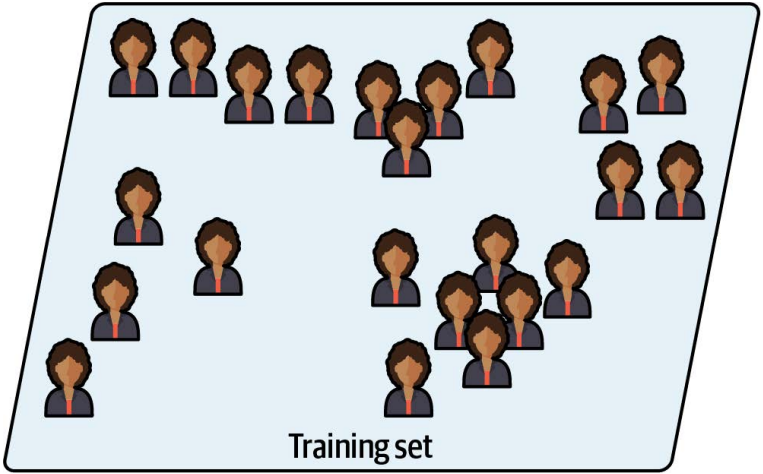
Supervised learning – algorithms

Here are some of the most important supervised learning algorithms that we will cover:

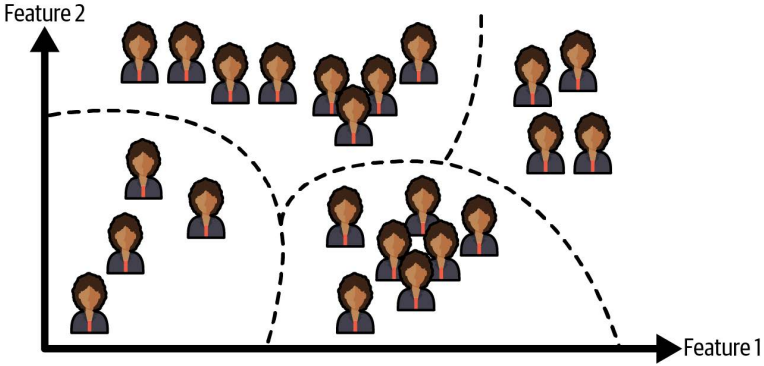
- Linear Regression
- Logistic Regression
- k-Nearest Neighbors
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests

Unsupervised learning

The training data is *unlabelled*.

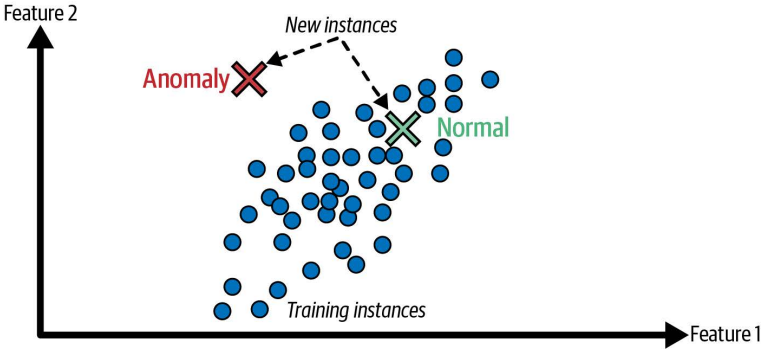


Unsupervised learning – clustering



Unsupervised learning – anomaly detection

Objective is to learn what "normal" data looks like, and then use that to detect abnormal instances.



Unsupervised learning

Here are some of the most important unsupervised learning algorithms:

- Clustering
  - k-means clustering
  - Density-based spatial clustering of applications with noise (DBSCAN)
  - Hierarchical Cluster Analysis (HCA)
- Visualization and dimensionality reduction
  - Principal Component Analysis (PCA)
  - Locally-Linear Embedding (LLE)
  - t-distributed Stochastic Neighbor Embedding (t-SNE)
- Association rule learning
  - Apriori algorithm
  - Eclat algorithm

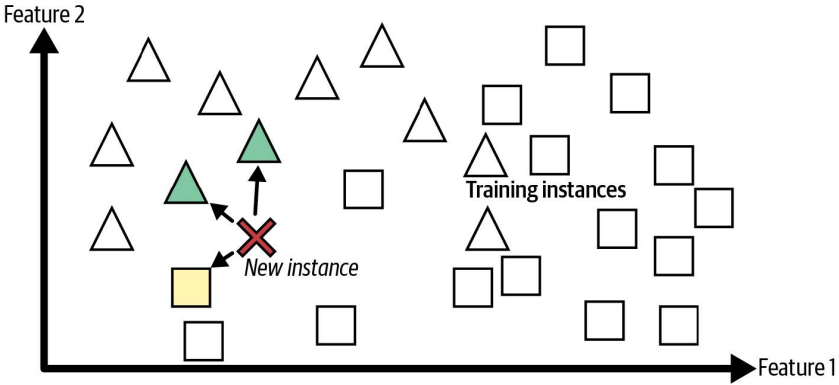
Instance-based learning vs model-based learning

Another way to categorise Machine Learning systems is by how they generalise. Most Machine Learning tasks are about making predictions.

This means that given a number of training examples, the system needs to be able to generalise to examples it has never seen before. Having a good performance measure on the training data is good, but insufficient; the true goal is to perform well on new instances.

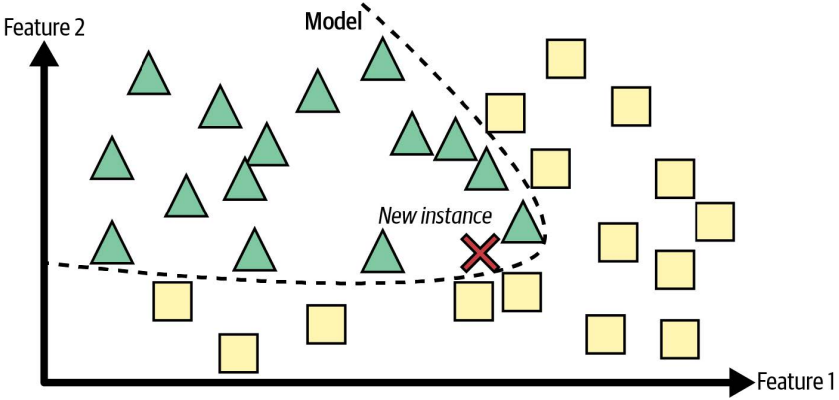
There are two main approaches...

Instance-based learning



"Memorise" the training examples, and generalise to new cases by using a similarity measure to compare to the learned examples.

Model-based learning



Build a model from the training examples, and use that model to make predictions.

Some ways to classify ML systems

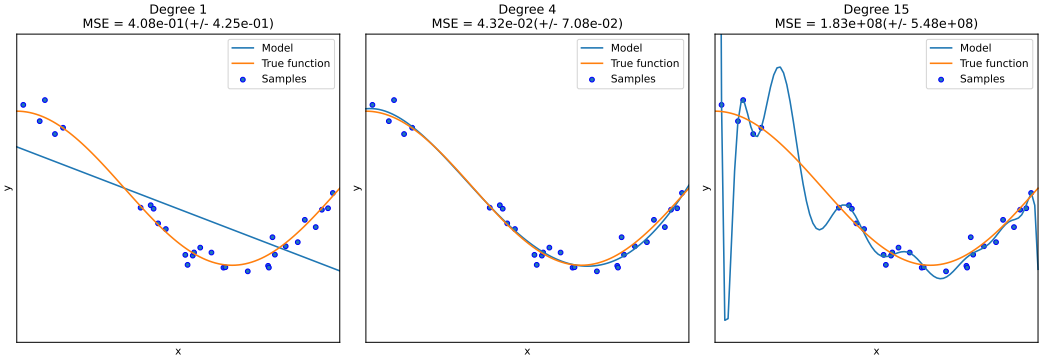
- Whether or not they can learn incrementally on the fly (online learning vs. batch learning)
- How they are supervised during training (supervised, semi-supervised, unsupervised, reinforcement learning).
- How they generalise (instance-based, model-based).

These criteria are not exclusive; you can combine them. We talked about some types today. Recommended reading includes studying about the other ones in the textbook.

Main challenges of machine learning

- Insufficient quantity of training data
- Non-representative or poor-quality data
- Irrelevant features
- Overfitting or underfitting

Overfitting and underfitting



Adapted from *"Underfitting vs. Overfitting,"* scikit-learn.
Available: https://scikit-learn/stable/auto_examples/model_selection/plot_underfitting_overfitting.html

Simple testing and validation

- Careful evaluation of the performance of ML models is very important.
- But how do we know if our model is any good?

Split your data into two sets: the training set and the test set. Train your model using the training set, and you test it using the test set.

- It is common to use 80% of the data for training and hold out 20% for testing.
- The error rate on the test set is called the *generalisation error* or *out-of-sample error*.

Because using the test set to pick the best *hyperparameter values* tends to make the model not perform well new data other than the test set, we should also need a *validation set*.

Summary

- ML is about making machines get better at some task by learning from data, instead of having to explicitly code rules.
- Many different types of ML systems: **supervised** or **unsupervised**; **batch** or **online**; **instance-based** or **model-based**, ...
- In a ML project you gather data in a training set, and you feed the training set to a learning algorithm. If the algorithm is model-based it tunes some parameters to fit the model to the training set (i.e., to make good predictions on the training set itself), and then hopefully it will be able to make good predictions on new cases as well. If the algorithm is instance based, it just learns the examples by heart and uses a similarity measure to generalise to new instances.
- The system will not perform well if your training set is too small, or if the data is not representative, noisy, or polluted with irrelevant features. Your model needs to be neither too simple nor too complex.